THERMAL SIMULATION REPORT

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1. Simulation Results Summary

Status	Success
Convergence	Yes
Calculated Heatsink Rth	0.0500 K/W
Max Base Temperature	85.50 °C
Max Junction Temperature	110.20 °C (Mod2_IGBT1)
Max NTC Temperature	90.00 °C
Outlet Air Temperature	55.00 °C

2. Input Parameters

Heatsink Geometry & Material

Dimensions (Lx x Ly)	0.25 m x 0.35 m
Base Thickness	0.02 m
Conductivity	218.0 W/mK
Rth Mode	Auto-Calculated (Fins)
Fins	31 fins (h=0.112m, t=0.005m)

Environmental Conditions

Inlet Air Temp	40.0 °C
Air Flow Rate	750.0 m³/h

3. Module Configuration

Module ID	Position (x, y) [m]	Power Losses [W]
Mod1	(0.1, 0.1)	IGBT1:100, Diode1:50
Mod2	(0.2, 0.2)	IGBT1:120, Diode1:60

4. Visualizations